TestBoard_ADC124
Grouped MUX timing and SPI engines

13 October 2026 | scanorder mux throughout | Production rates | Manual / repeat 1

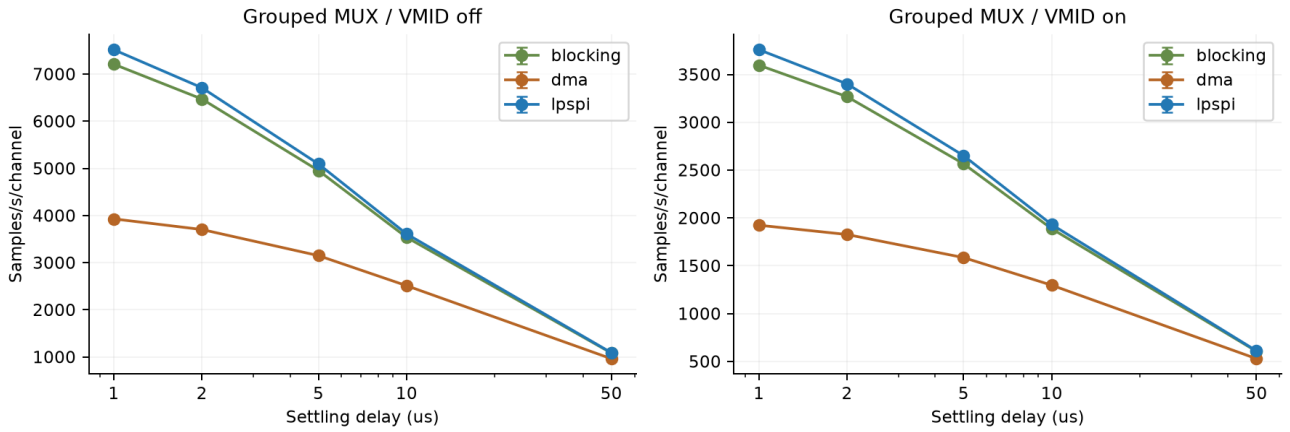
The intended address-grouped scan

Every production and diagnostic capture uses scanorder mux. Switch and settle once at each external address; read ADC IN1-IN4 for addresses 0-5, then IN4 alone at address 6. Complete array A, park it, complete array B and park it. There are 25 useful values per array, 50 per full sweep. This report replaces the earlier channel-order primary timing comparison for the requested scan plan.

Both connectors disconnected; charger connected. Bias resistors A: 470 kOhm / 1 MOhm / 470 kOhm / 1 MOhm; B: 1 MOhm / 470 kOhm / 1 MOhm / 470 kOhm. Nominal reference 3.3 V, VMID 1.65 V. Requested SPI clock 16 MHz on one shared bus; the two ADCs acquire sequentially.

Production sampling rate: three repetitions

Settle us	VMID	Blocking Hz/ch	DMA Hz/ch	LPSPI Hz/ch	LPSPI vs blocking	LPSPI vs DMA
1	off	7,210.9	3,929.3	7,518.1	4.3%	91.3%
1	on	3,597.1	1,924.2	3,759.4	4.5%	95.4%
2	off	6,472.0	3,703.7	6,711.4	3.7%	81.2%
2	on	3,268.0	1,827.1	3,401.4	4.1%	86.2%
5	off	4,955.6	3,150.4	5,094.1	2.8%	61.7%
5	on	2,566.4	1,585.7	2,651.3	3.3%	67.2%
10	off	3,537.7	2,512.6	3,610.1	2.0%	43.7%
10	on	1,886.8	1,297.0	1,930.5	2.3%	48.8%
50	off	1,084.8	963.9	1,091.4	0.6%	13.2%
50	on	606.2	528.9	610.8	0.8%	15.5%



90 fresh production captures: 30 settings x 3 repetitions, each 1 s warm-up + 15 s measured. Configuration order rotates between repetitions. Rates use device sweep-start timestamps after warm-up. Table entries are repetition medians; error bars show the min-max range. Useful total values/s = 50 x Hz/channel.

Operations per address, per ADC and per sweep

Address per array	Groups / array	Useful values / group	Words / group off	Words / group on
0-5	6	4	5	10
6 / IN4 alone	1	1	2	7

Full two-array sweep	VMID off	VMID on
Sensor address groups	14	14
Actual sensor switches	14	14
Additional VMID park switches	0	14
Additional mandatory park changes	2	0 (calls return already parked)
Total physical address changes	16	28
Selector-prime sensor words	14	14
Retained sensor words	50	50
Discarded VMID words	0	70
Total 16-bit ADC words	64	134

A sensor group includes one selector prime plus its 4 (or 1) retained conversions. VMID-on adds five discarded words per group: one selector prime plus all four ADC inputs. End/handoff parking still runs; with insertion enabled those calls do not produce another address change or settling wait.

Each real external-address change performs EN off, a 200 ns allowance, three address GPIO writes, EN on, a 200 ns allowance and the configured extra analog settling. Each ADC word has 100 ns CS setup and 100 ns CS-high allowance in the current driver. GPIO/API overhead and interrupts add to these explicit waits.

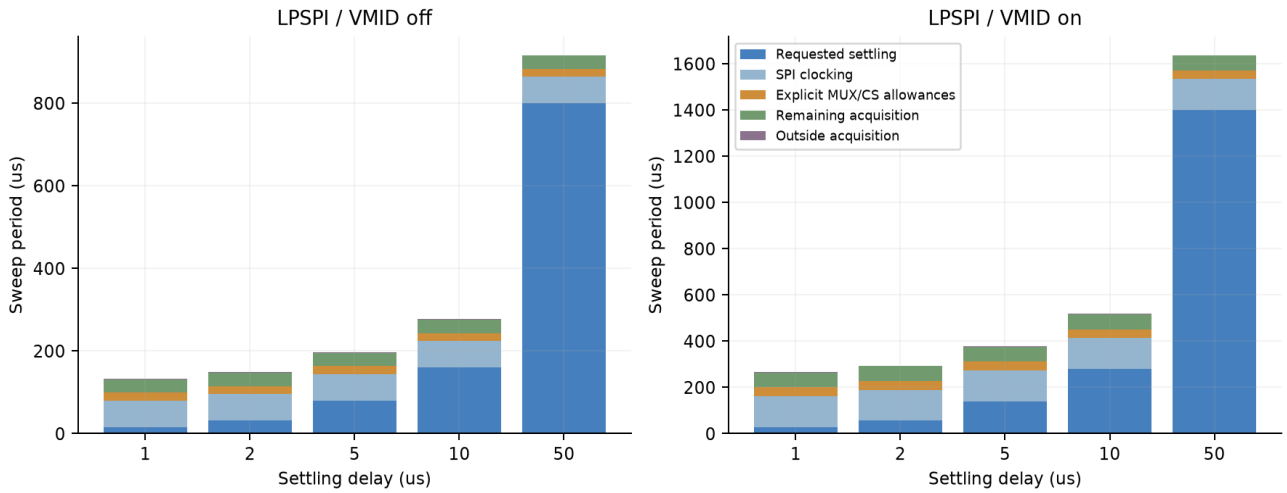
At nominal 16 MHz, one 16-clock ADC frame takes 1.000 us: 3 clock cycles of track/acquisition (0.1875 us), followed by 13 conversion/output cycles (0.8125 us). Conversion and output shifting occupy the same frame, so they are not added again as independent costs. The DIN selector applies to the next conversion. TI ADC124S101 datasheet, Operation / Using the ADC.

The pinned SPI library selects a 240 MHz peripheral root and divider 15 for the 16 MHz request. Calculations use this nominal frequency; electrical GPIO propagation and the SCLK waveform were not independently scoped.

LPSPI production timing budget

Period = acquisition + outside-acquisition gap. Acquisition is split into requested settling, nominal SPI clocking, explicit MUX/CS allowances and a remaining cost. The remaining cost contains GPIO/register work, driver loops, delay-call overhead and interrupts; it is not a uniquely measured phase.

Settle us	VMID	Settle us	SPI clock us	Explicit waits us	Remaining acquisition us	Gap us	Period us
1	off	16.00	64.00	19.20	32.52	1.29	133.01
1	on	28.00	134.00	38.00	64.82	1.18	266.00
2	off	32.00	64.00	19.20	32.14	1.66	149.00
2	on	56.00	134.00	38.00	64.95	1.05	294.00
5	off	80.00	64.00	19.20	32.00	1.10	196.30
5	on	140.00	134.00	38.00	64.08	1.10	377.18
10	off	160.00	64.00	19.20	32.48	1.32	277.00
10	on	280.00	134.00	38.00	64.94	1.06	518.00
50	off	800.00	64.00	19.20	31.99	1.11	916.29
50	on	1400.00	134.00	38.00	64.08	1.13	1637.21



Off: settling = 16 x delay; clocking = 64 us; explicit allowances = 16 x 0.4 + 64 x 0.2 = 19.2 us. On: settling = 28 x delay; clocking = 134 us; explicit allowances = 28 x 0.4 + 134 x 0.2 = 38.0 us. The residual closes the measured budget. Rows are medians of repetition means; independently summarized rounded components may differ slightly from their displayed sum.

Rate-versus-settling fits and engine overhead

Fit full sweep period T = fixed intercept + slope x settling delay. A slope near 16 off or 28 on validates the number of actual waits. The intercept is an extrapolated combined fixed cost, including transfer/GPIO/loop/USB work; it is not just ADC conversion time.

Engine	VMID	Measured slope us/us	Expected waits	Intercept us	Max fit residual us
blocking	off	15.9875	16	122.466	0.612
blocking	on	27.9952	28	249.935	0.257
dma	off	15.9875	16	238.045	0.558
dma	on	27.9869	28	491.233	0.550
lpspi	off	15.9884	16	116.880	0.518
lpspi	on	27.9872	28	237.851	0.609

Per-channel amortized production cost

Settle us	VMID	Acquisition / useful value us	Full period / useful value us	50-channel useful values/s
1	off	2.6345	2.6602	375,907
1	on	5.2965	5.3200	187,970
2	off	2.9469	2.9800	335,570
2	on	5.8591	5.8800	170,068
5	off	3.9041	3.9261	254,707
5	on	7.5216	7.5436	132,563
10	off	5.5136	5.5400	180,505
10	on	10.3388	10.3600	96,525
50	off	18.3038	18.3259	54,568
50	on	32.7216	32.7442	30,540

These are shared-work costs divided by 50, not the ADC conversion duration or individual sample apertures. Four ADC inputs at one external address are read sequentially. Exact per-channel timestamps are not in the frame; each value shares the sweep start/end interval.

Measured diagnostic phases

30 fresh grouped-MUX diagnostic captures: one repetition per setting, 1 s warm-up + 2 s measurement. Existing profile firmware records CPU cycles at 600 MHz. Means include warm-up, interrupts and probe overhead; production rate values are kept separate. Table shows 5 us settling.

Engine	VMID	Prepare us	Session us	Transfer us	Validation us	Encode us	USB enqueue us	Profiler book-keeping us
blocking	off	92.939	0.027	112.004	0.664	0.242	0.414	1.842
blocking	on	81.477	0.027	314.581	1.413	0.242	0.415	1.934
dma	off	93.402	0.027	227.768	0.762	0.242	0.422	1.942
dma	on	81.690	0.027	557.405	1.585	0.243	0.420	1.991
lpspi	off	93.928	0.442	103.986	0.757	0.242	0.413	1.917
lpspi	on	81.402	0.439	298.540	1.573	0.242	0.413	1.893

Nested scopes: validation is inside transfer; prepare/session/transfer are inside acquisition. Transfer with VMID on includes the additional VMID parking and settling. Do not sum every phase column. Profiler bookkeeping is extra diagnostic work absent from the ordinary production build.

Settle us	VMID	Production period us	Diagnostic period us	Probe/build slowdown
1	off	133.012	138.063	3.80%
1	on	266.000	273.000	2.63%
2	off	149.000	154.045	3.39%
2	on	294.000	301.048	2.40%
5	off	196.304	203.000	3.41%
5	on	377.178	385.000	2.07%
10	off	277.000	282.672	2.05%
10	on	518.000	525.158	1.38%
50	off	916.295	921.920	0.61%
50	on	1637.208	1644.503	0.45%

The slowdown is a whole-build comparison. Adjacent cycle-read calibration cannot remove every probe and bookkeeping operation, so individual phases are not presented as corrected production timings.

Per-address scan-time estimates

For each engine, off transfer / 64 gives an aggregate cost per conversion word, including its share of the 14 read-batch calls and profiling overhead. Prepare residual / 16 subtracts requested settling and includes the 0.4 us switch allowances plus GPIO/delay/probe work. With only one topology these measurements cannot independently separate word and batch-call overhead.

Engine	Aggregate transfer / word us	Prepare residual / actual switch us
blocking	1.7359	0.8087
dma	3.5564	0.8250
lpspi	1.6282	0.8061

The following 5 us LPSPI estimates use uniform aggregate word cost. Scan includes all priming/retained/VMID words; switch/settle includes the group's sensor address change and any VMID park. Shared mandatory parking, session setup, outer-loop and USB costs are excluded. A particular group's driver overhead may differ from its allocated average.

Group per ADC	VMID	Values	Words	Switch us	Settle us	Scan us	Service us	Service / value us
Addresses 0-5	off	4	5	0.806	5.0	8.141	13.947	3.487
Addresses 0-5	on	4	10	1.612	10.0	16.282	27.894	6.973
Address 6 / IN4	off	1	2	0.806	5.0	3.256	9.062	9.062
Address 6 / IN4	on	1	7	1.612	10.0	11.397	23.009	23.009

Each ADC has six four-value groups and one single-value group. VMID-off adds one mandatory park per array; the two-array sweep has two. VMID-on parking is already included at every group. Per-channel service is amortized, not simultaneous: IN1-IN4 conversion results are acquired in sequence.

USB work and optimization priorities

Settle us	VMID	Production gap us	Profile encode us	Profile enqueue us	USB kB/s	Gap share
1	off	1.287	0.242	0.413	857.1	0.97%
1	on	1.175	0.242	0.413	428.6	0.44%
2	off	1.657	0.242	0.413	765.1	1.11%
2	on	1.046	0.242	0.413	387.8	0.36%
5	off	1.099	0.242	0.413	580.7	0.56%
5	on	1.100	0.242	0.413	302.2	0.29%
10	off	1.319	0.242	0.413	411.6	0.48%
10	on	1.059	0.243	0.413	220.1	0.20%
50	off	1.106	0.243	0.413	124.4	0.12%
50	on	1.128	0.243	0.415	69.6	0.07%

A frame contains 114 bytes: 4-byte header, 100-byte values, 10-byte timing trailer. USB CDC is not limited by the configured 460800 UART baud value. The core partial-buffer flush timer is nominally 75 us and asynchronous; it is not an added sampling-loop wait. Enqueue measures a queue copy, not physical USB transmission or host receipt. External transport latency and negotiated USB speed were not measured.

What can still improve the intended grouped scan

Engine: use the measured engine ranking above. LPSPI keeps one SPI transaction per sweep; blocking sets up/ends a transaction per word, while current DMA starts/completes an asynchronous two-byte transfer per word. The grouped scan is already active throughout these results, so address grouping is not presented as an additional future optimization.

Burst within an address: keeping CS low for the prepared 5/2-word sensor batches and 5-word VMID batches could reduce per-word CS/API/DMA overhead. Pure explicit CS allowance savings would be (64-14) x 0.2 = 10.0 us/sweep off and (134-28) x 0.2 = 21.2 us/sweep on. Software savings are unmeasured. TI supports multiple back-to-back 16-clock conversions with CS held low; a new driver would still require timing/state/error and analog validation. TI serial-interface specification.

GPIO and settling: fast/prepared GPIO writes may reduce the fixed switch cost, preserving EN polarity and disable/enable allowances. Lower settling has a larger gain at long delays but must be selected from analog baseline/response evidence. Quiet open-connector data do not validate stimulated ghosting or attached sensors.

USB: the measured outside gap bounds how much period can be saved by encoding/enqueue/main-loop changes; USB interrupts also occur inside acquisition. Accounted busy discards quantify transport pressure. Prioritize USB work only if it materially limits delivery, rather than treating the asynchronous flush timer as a sampling delay.

Integrity and evidence

120 fresh grouped-MUX captures: 90 production and 30 diagnostic. Received 4,734,190 full sweeps (236,709,500 channel values including warm-up); acquired 4,734,291, deliberately discarded 101 whole sweeps. Every transmitted sweep was received; acquired equals sent plus discarded. Firmware error counters remained zero.

Every capture preserved raw bytes, command/status logs and SHA-256 hash. Firmware configuration and exact payload_routes were validated before run. A structured reader and existing incremental parser agreed on every value/timestamp/count, including reads split through frame boundaries. Codes stayed within 0-4095; wrapped timestamps were unwrapped, monotonic and non-overlapping. Hashes were rechecked for this report.

Ordinary production firmware was used for throughput. The existing diagnostic build was loaded for phase measurement with one CLI uploader and direct USB reboot, then production restored. Final status confirms profiling unavailable, running false and the original full-array/MUX-order/blocking/16 MHz/repeat-1/5 us/VMID-off configuration. No firmware acquisition source or protocol changes were made.

Quantity	Evidence / limit
Sweep period and delivered rate	Production device sweep-start intervals after warm-up
Acquisition and outside gap	Correctly paired production start/end intervals
Switch/word/group counts	Firmware plan and exact grouped payload routes
Settling and explicit wait budgets	Event counts x configured waits
Prepare/transfer/session/encode/enqueue	Cycle-counter diagnostic scopes; probes included
Per-address service estimates	Uniform allocation from aggregate word/switch costs
Exact electrical MUX/SCLK edge timing	Not independently measured with a logic analyzer
Individual channel apertures and USB host latency	Not transmitted/instrumented in these captures

CSV evidence: 90 production capture rows, 30 production settings, 30 diagnostic rows/settings, 6 period fits and 3 aggregate-unit rows. Timing_analysis.json includes original/final configuration and firmware image/source hashes. The companion grouped-MUX settling/VMID report contains quiet masks and full per-channel noise/baseline values in counts and volts.